Full Stack AI Engineering | 3-Month Program

Neshama
Agentic AI

STACK

Python + MERN + GenAI + Agentic AI

Multi Agentic Architecture



- Live + Recorded
- Real-world AI projects + Agentic systems
- Career preparation from Week 1

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WEBSITE www.neshamatraining.com

Beyond technical skills

Professional readiness is part of the curriculum

Building industry-ready engineers who take ownership and deliver real outcomes.

AI engineering requires more than technical ability. Learn to take ownership, communicate with clarity, collaborate effectively and make informed decisions.

LEARN THROUGH PRACTICAL EXPERIENCE

01

Workplace Simulations

Practice professional judgment in realistic engineering scenarios.

02

Collaborative Exercises

Work with others, align on priorities and communicate decisions clearly.

03

Project-Based Learning

Build AI applications while taking responsibility for delivery and outcomes.

- Move from completing assignments to owning meaningful engineering outcomes.

Program blueprint

3-Month Full-Stack AI Engineer Program. A 12-week journey built around recorded foundations, live AI engineering, real projects and career preparation.

PROGRAM MODEL

3 months / 12 weeks

Live + Recorded

Python + MERN + GenAI + Agentic AI

Project-led learning

RECORDED FOUNDATIONS

- Python fundamentals
- HTML / CSS / JavaScript
- React
- Node / Express / MongoDB

LIVE AI ENGINEERING

- LLMs + Prompt Engineering
- RAG + Vector Databases
- Agents + MCP + Coding Agents
- AI product architecture + deployment

PLACEMENT PROGRAM

- LinkedIn + GitHub + Portfolio
- Resume + Mock Interviews
- Corporate Readiness
- Recruiter outreach

Note Live time focuses on AI Engineering, product building and agentic systems. Python and MERN foundations remain available as recorded learning.

12-week roadmap

Three months. Eight modules. A parallel career track.

Month 01

LLM & AI Application Engineering

- WK 01 Engineer Setup + Career Foundation
- WK 02 Generative AI & LLM Fundamentals
- WK 03 Prompt Engineering + LLM APIs
- WK 04 AI-Assisted Development + AI Copilot

Month 02

RAG & AI Product Engineering

- WK 05 Embeddings + Vector Databases
- WK 06 RAG + Advanced RAG
- WK 07 FastAPI + React + AI Product Architecture
- WK 08 Full-Stack AI Product + Deployment

Month 03

Agentic AI + Production + Career

- WK 09 Agent Fundamentals + Tool Calling
- WK 10 Codex + Claude Code + Hermes + Grok + MCP
- WK 11 Multi-Agent Systems + Memory + Evaluation
- WK 12 Capstone + Resume + Mock Interviews + Career Readiness

Module 01

RECORDED FOUNDATION

Python Foundations

OUTCOME

Build a solid Python base and begin using AI as a development assistant.

RECORDED CORE

- Python syntax, variables and data types
- Conditional statements and loops
- Lists, tuples, sets and dictionaries
- Functions, modules and packages
- OOP fundamentals
- File handling and exception handling
- JSON, HTTP and API basics
- Git and GitHub fundamentals

AI-ASSISTED CODING

- Using AI to explain code
- AI-assisted debugging
- Code generation and refactoring
- Structured AI outputs
- Introduction to LLM API usage

MODULE ASSIGNMENTS

AI Quiz Studio

Generate structured quizzes from natural-language topics.

Code Insight

Analyze source code and generate AI-powered explanations.

Module 02

RECORDED FOUNDATION

Web & MERN Foundations

OUTCOME

Build the web foundation needed to ship AI-powered interfaces.

FRONTEND

- HTML and semantic structure
- CSS layout and responsive design
- Modern JavaScript
- Promises and Async/Await
- Fetch API and browser APIs
- React components, props, state and hooks
- Routing and forms

BACKEND

- Node.js fundamentals
- Express.js and REST APIs
- MongoDB and data modeling
- Authentication basics
- Frontend/backend integration
- Streaming responses in the browser

PROJECT DELIVERABLES

AI Productivity Board
AI-assisted task and productivity workflow.

Smart Content Desk
Generate, rewrite, summarize and organize content.

Module 03

LIVE AI ENGINEERING

LLM Engineering

OUTCOME

Build production-style AI features on top of modern LLM APIs.

WEEK 1–2 TOPICS

- Generative AI fundamentals
- LLM architecture
- Tokens and tokenization
- Context windows
- Transformers and attention
- Parameters and inference
- Temperature, Top-P and output limits
- Hallucinations and model limitations

PROMPT ENGINEERING

- System and user prompts
- Zero-shot and few-shot prompting
- Role prompting
- Prompt templates
- Structured JSON outputs
- Prompt optimization
- Prompt evaluation

LLM APIS

- OpenAI API
- Anthropic / Claude API
- Google Gemini API
- Grok API concepts
- OpenRouter
- Streaming responses
- Function calling
- Error handling, rate limits and cost awareness

GEN AI AND AGENTIC PROJECTS

Neshama AI Copilot
Multi-model assistant with streaming and configurable instructions.

Prompt Lab
Experiment, compare and evaluate prompt outputs.

Module 04

LIVE AI ENGINEERING

RAG & Knowledge Engineering

OUTCOME

Build AI systems that can answer questions using private or domain-specific information.

RETRIEVAL FOUNDATIONS

- Embeddings
- Semantic similarity
- Vector representations
- FAISS
- ChromaDB
- Pinecone
- Indexing and metadata
- Similarity retrieval

RAG PIPELINE

- Document ingestion
- Document processing
- Chunking strategies
- Embedding generation
- Vector storage
- Retrieval
- Context construction
- Prompt assembly and answer generation

ADVANCED RAG

- Metadata filtering
- Query rewriting
- Hybrid search
- Reranking
- Context optimization
- Retrieval evaluation
- Groundedness and hallucination reduction

PROJECT DELIVERABLES

Knowledge Vault
Document upload, embeddings, retrieval and grounded answers.

Business Knowledge Assistant
Domain-specific RAG for organizational information.

Module 05

LIVE AI PRODUCT DEVELOPMENT

Full-Stack AI Product Engineering

OUTCOME

Turn individual AI capabilities into complete software products.

BACKEND & APPLICATION

- Python + FastAPI
- REST API architecture
- Request validation
- Authentication and JWT
- File uploads
- Database integration
- AI service layer
- Error handling and configuration

FRONTEND & PRODUCT

- React integration
- AI chat interfaces
- Streaming UI
- Loading and error states
- Conversation history
- User flows
- Product architecture
- Frontend > Backend > AI model integration

SHIPPING FOUNDATIONS

- Git and GitHub
- Environment variables
- Secrets management
- Application configuration
- Basic production architecture

PROJECT DELIVERABLE

AI Operations Workspace

React + Python + database + LLM/RAG full-stack product.

Module 06

LIVE AGENT SYSTEMS

Agentic AI Engineering

OUTCOME

Design agents that can reason through tasks, use tools and execute multi-step workflows.

AGENT FUNDAMENTALS

- What is Agentic AI?
- Assistant vs Agent
- Perception > Reasoning > Action
- Planning
- Task decomposition
- Reasoning loops
- Tool selection and execution
- State management
- Short-term and long-term memory

AGENT DESIGN PATTERNS

- Single-agent systems
- Tool-using agents
- Router agents
- Planner-Executor agents
- Reflection agents
- Sequential workflows
- Parallel workflows
- Human-in-the-loop patterns

AGENT FRAMEWORKS

- LangChain
- LangGraph
- LlamaIndex
- Agent state
- Graph-based workflows
- Conditional routing
- Loops and orchestration
- Multi-step execution

PROJECT DELIVERABLES

ResearchPilot
Autonomous research workflow using tools, planning and structured outputs.

Workflow Agent
Multi-step business workflow executor.

Module 07

LIVE AGENTIC DEVELOPMENT

AI Coding Agents & MCP

OUTCOME

Build agentic developer workflows and connect AI systems to real tools.

AI CODING AGENTS

- OpenAI Codex
- Claude Code
- Hermes Agent
- Grok
- Cursor
- GitHub Copilot

- Repository understanding
- Task delegation
- Code generation
- Debugging and refactoring
- Test generation
- Code review
- Git workflows

AGENTIC CODING WORKFLOW

- Requirement > plan > implementation
- Run tests and inspect results
- Iterative debugging
- Review generated changes
- Prepare commits and documentation
- Human approval and permissions

MCP

- Model Context Protocol fundamentals
- MCP clients and servers
- Tools
- Resources
- Prompts
- Custom MCP servers
- Connecting agents to databases and APIs
- MCP with coding agents and custom applications

PROJECT DELIVERABLES

DevOps Copilot
Repository-aware coding agent that inspects, modifies and tests code.

MCP Command Center
Connect an AI agent to external tools, APIs and data.

Module 08

LIVE + CAPSTONE

Production AI Engineering

OUTCOME

Deploy reliable AI systems and understand the operational concerns behind production agents.

PRODUCTION AGENT SYSTEMS

- Multi-agent systems
- Supervisor and worker agents
- Agent communication
- Memory architecture
- Persistent state
- Task delegation
- Result aggregation

EVALUATION & SAFETY

- Agent reliability
- Tool errors
- Infinite-loop prevention
- Prompt injection awareness
- Tool injection awareness
- Logging and tracing
- Cost monitoring
- Evaluation datasets
- Automated evaluation

DEPLOY & OPERATE

- Vercel and backend deployment
- Environment configuration
- API security
- Rate limiting
- Token optimization
- Caching
- Monitoring
- Production readiness

PROJECT DELIVERABLE

Autonomous Business Agent

RAG + tools + memory + multi-agent workflow.

A dedicated engineering track

Agentic AI deep dive

Build agents that plan, select tools, execute tasks, inspect results and return verified outputs.

01

Agent Architecture

- Planning + reasoning loops
- State + memory
- Tool selection
- Task decomposition

02

Tools & Actions

- Function calling + APIs
- Databases + search
- Files + code execution
- Tool validation

03

Orchestration

- LangChain + LangGraph
- LlamaIndex
- Sequential + parallel flows
- Router patterns

04

Coding Agents

- Codex + Claude Code
- Hermes Agent + Grok
- Cursor
- GitHub Copilot

05

MCP

- Clients + servers
- Tools + resources
- Prompts
- Custom MCP servers

06

Multi-Agent Systems

- Supervisor + worker agents
- Research + coding agents
- Reviewer agent
- Result aggregation

AI tools & technology stack

Tools taught through practical workflows

AI MODELS & APIS	OpenAI · Claude / Anthropic · Gemini · Grok · OpenRouter
CODING AGENTS	Codex · Claude Code · Hermes Agent · Cursor · GitHub Copilot
AI FRAMEWORKS	LangChain · LangGraph · LlamaIndex · Hugging Face
RAG & VECTOR	Pinecone · FAISS · ChromaDB · Embeddings
FULL-STACK	Python · FastAPI · JavaScript · React · Node.js · Express · MongoDB · SQL
SHIP & DEPLOY	Git · GitHub · Postman · Swagger / OpenAPI · Docker · Vercel · AWS
LOCAL AI	Ollama · Open-source models · Local inference

Career track from Week 1

Professional presence grows with technical skills



WEEK 1

Profile Foundation

- LinkedIn headline and About section
- Skills and featured section
- GitHub profile README
- Portfolio structure

WEEKS 2–4

Build the Signal

- Commit foundation work
- Document first AI projects
- Publish learning / project updates
- Add projects to portfolio

WEEKS 5–8

Showcase AI Engineering

- RAG project case study
- Architecture diagrams + live demos
- Technical LinkedIn posts
- GitHub repository cleanup

WEEKS 9–10

Demonstrate Agentic Skills

- Agent project write-up
- MCP workflow documentation
- Coding-agent workflow showcase
- Demo video / walkthrough

WEEKS 11–12

Get Interview Ready

- AI Engineer resume + ATS-oriented project descriptions
- Mock interviews + project defense
- Recruiter outreach + job search workflow

AI-Native Project Portfolio | 01–06

AI is the core product layer

From model APIs and grounded retrieval to agentic workflows and production systems.

01

AI Copilot

LLM APIs + Streaming

Multi-model assistant with configurable instructions.

02

Prompt Lab

Prompt Engineering

Test, compare and evaluate model outputs.

03

AI Productivity Board

MERN + AI

AI-assisted tasks and productivity workflows.

04

Knowledge Vault

Embeddings + RAG

Chat with uploaded documents using grounded retrieval.

05

Business Knowledge Assistant

Advanced RAG

Answer domain-specific business questions using private knowledge.

06

AI Operations Workspace

Python + React + DB

Complete full-stack AI product with secure APIs.

AI-Native Project Portfolio | 07–12

AI is the core product layer

From model APIs and grounded retrieval to agentic workflows and production systems.

07

ResearchPilot

Agents + Tools

Autonomous research agent with planning and tool use.

08

Workflow Agent

Agents + APIs

Execute multi-step operational workflows.

09

DevOps Copilot

Coding Agents

Repository-aware coding, testing and review workflow.

10

MCP Command Center

MCP + Agents

Connect AI agents to external tools and systems.

11

Multimodal AI Studio

Vision + Audio + LLM

Combine text, image and audio AI capabilities.

12

Autonomous Business Agent

RAG + Multi-Agent

Business assistant with memory, tools and verification.

High-signal communication

Professional communication in everyday engineering

Take ownership of outcomes, communicate clearly and keep stakeholders aligned. Learn to communicate like an engineer who owns outcomes. Make progress visible, identify risks early and turn updates into action.

PRACTICE REAL WORKPLACE COMMUNICATION

- Standups
- Stakeholder updates
- Project presentations
- Professional communication exercises

01	**Clarity**	Explain progress, challenges and decisions without ambiguity.
02	**Brevity**	Share meaningful updates without unnecessary detail.
03	**Actionability**	Make next steps, owners and required decisions explicit.
04	**Accountability**	Identify risks early, escalate blockers and propose solutions.
05	**Stakeholder Alignment**	Adapt your message for teammates, managers and nontechnical stakeholders.

Interview-ready roles

Software engineering + AI application delivery

01
AI Engineer

- LLM applications
- RAG systems
- Agents + AI integrations

02
Full-Stack AI Developer

- React + Python
- AI backends + interfaces
- Deployment

03
LLM Application Developer

- Model APIs
- Prompt engineering
- Function calling + streaming

04
AI Application Engineer

- AI integration + APIs
- Data + AI workflows
- Production readiness

05
AI Automation Engineer

- Agents + tools
- MCP
- Workflow automation

06
AI Product Engineer

- AI feature design
- Full-stack delivery
- Architecture + shipping

Mock interviews & corporate readiness

Practice alongside the technical curriculum

Technical Interviews

- Python + JavaScript / React
- APIs + databases
- LLMs + RAG
- Agents + MCP

AI Engineering Discussion

- Explain AI architecture + RAG
- Explain vector search
- Explain agent workflows + model choices
- Discuss production trade-offs

Mock Interviews

- Technical mock interview
- AI engineering mock
- Project defense
- HR / communication practice

Corporate Readiness

- Professional communication
- Email etiquette + meetings
- Team collaboration + Agile workflow
- Technical presentation

- Career preparation runs in parallel throughout the program, with concentrated interview practice in the final weeks.

What students finish with

A practical portfolio and a professional profile

01
Python + MERN foundation access

02
LLM application development experience

03
Prompt engineering capability

04
Working RAG system

05
Vector database experience

06
AI Agent and multi-agent experience

07
MCP and tool-integration experience

08
Hands-on exposure to Codex, Claude Code, Hermes Agent and Grok



DELIVERABLES

- Multiple GitHub repositories
- Portfolio-ready case studies
- LinkedIn profile developed from Week 1
- Live-deployed AI application
- AI Engineer resume
- Mock interview experience
- Corporate readiness preparation
- Certificate of completion

Program at a glance

Build, Deploy & Get Interview-Ready

DURATION
3 Months / 12 Weeks

MODE
Live + Recorded

FOUNDATION
Python + MERN recorded content

LIVE FOCUS
LLMs, Prompt Engineering, RAG, Agents, MCP, AI Products, Deployment

AGENTIC STACK

LangChain | LangGraph | LlamaIndex | Codex | Claude Code | Hermes Agent | Grok | MCP

CAREER TRACK

LinkedIn + GitHub + Portfolio from Week 1; Resume + Mock Interviews + Corporate Readiness

PROJECT PORTFOLIO

AI Copilot, RAG system, Full-Stack AI product, Research Agent, Coding Agent, MCP system, Autonomous Business Agent

START YOUR AI ENGINEERING JOURNEY

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Meet your course lecturers

Learn from people who build, ship and run software in industry

VP

FOUNDER & CTO

Venkata Prithvi Krishna Palle

Founder & CTO @ Vaulth AI

Turning AI ideas into products people use.

- Founder & CTO of Vaulth AI
- Leads AI product strategy and engineering
- Designs production-grade AI systems end to end
- Builds and mentors engineering teams
- Teaches the architecture behind real AI products

AI Products | Engineering Leadership | AI Systems

M

25+ YEARS

Murthy

Product Manager · 25+ years of experience in software

Three decades of shipping software, shared.

- Product Manager with 25+ years in software
- Has guided products from idea to release
- Deep experience in software delivery
- Bridges business needs and engineering teams
- Brings real industry context to every lesson

Product Management | Software Delivery | Industry Experience

VM

DATAOPS & PYTHON

Vamshi Maraboina

Senior Analyst · DataOps Automation Engineer · Python Developer

Automating data, one pipeline at a time.

- Senior Analyst and DataOps Automation Engineer
- Python developer building data workflows
- Automates pipelines for reliable data delivery
- Hands-on with testing, scripting and monitoring
- Guides learners through Python foundations
- Experienced in GenAI, LangChain and LangGraph and AI Agentic systems

Python | DataOps | Automation | GenAI | LangGraph

Program fees

Your next step towards AI engineering

All prices are inclusive of taxes.

Core Program

3-Month Full-Stack AI Engineering

Program fee

Rs. 25,000

- Python + MERN + GenAI + Agentic AI
- ~~Multi Agentic Architecture~~
- ~~Placement assistance~~
- ~~Corporate training~~

Complete Program

3-Month Full-Stack AI Engineering

Program fee · ~~Rs. 1,00,000~~

Rs. 55,000

SAVE Rs. 45,000 | 45%

- Python + MERN + GenAI + Agentic AI
- Multi Agentic Architecture
- Placement assistance
- Corporate training

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